VoxOps: A Unified Voice-First AI Platform for Customer Experience and IT Operations with Emotion-Aware Reinforcement Learning

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Abstract—Enterprise voice AI remains fragmented: customer support bots and IT operations tools operate as isolated silos despite sharing 73% of their underlying knowledge graphs. We present VoxOps, a unified voice-first AI platform that bridges customer experience (CX) and IT operations through a shared LLM reasoning backbone with domain-specific LoRA adapters, achieving sub-200 ms voice-to-voice latency via speculative LLM prefill on partial ASR transcripts. Our contributions include: (1) a Unified Voice Reasoning Architecture (VRA) where a single Llama-3.1-70B backbone serves both customer and operations contexts through hot-swappable LoRA adapters with cross-domain knowledge transfer; (2) a sub-200 ms streaming pipeline using speculative prefill—beginning LLM inference on partial ASR output (188 ms P50 voice-to-voice, below human perception threshold); (3) an emotion-aware RL conversation policy with a 12-action space that explicitly optimizes customer emotional trajectory, not just resolution speed (CSAT +18%, AHT −23%); (4) 25-language support with accent-robust ASR (4.1% WER on accented speech vs. 12.3% baseline) and culturally-adapted conversation policies; (5) autonomous operations remediation via 47 voice-executable runbooks with dual-confirmation safety; and (6) a voice-driven knowledge graph that captures tribal engineering knowledge from every conversation. Deployed across 2M+ conversations, VoxOps achieves 78% first-call resolution (vs. 52% baseline), 4.3/5.0 CSAT, and automates 67% of L1/L2 operations tasks.

Patent Notice: The speculative prefill pipeline and emotion-aware RL conversation policy are subject to pending patent applications.

Index Terms—Voice AI, conversational AI, emotion detection, reinforcement learning, IT operations, multilingual ASR, digital operations, knowledge graph

I. Introduction

Enterprise voice interactions span two critical domains: customer support (billing, provisioning, troubleshooting) and IT operations (alert triage, incident management, runbook execution). Today, these domains use entirely separate toolchains—Amazon Connect/Lex for customers, PagerDuty/OpsGenie for operations—despite sharing the same infrastructure knowledge base, incident history, and service topology.

This separation causes three failure modes: (1) Knowledge silos: When a customer reports "my service is slow," the support bot cannot access the operations knowledge graph showing an active maintenance window. (2) Duplicate investment: Both domains require ASR, NLU, dialog management, and TTS—built and maintained independently. (3) Latency: Current voice bots exhibit 600–1200 ms voice-to-voice latency, producing unnatural conversational pauses that degrade user satisfaction.

VoxOps addresses all three through a unified architecture with six technical innovations detailed in this paper.

II. Unified Voice Reasoning Architecture

A. Shared Backbone with LoRA Adapters

The core insight is that customer support and IT operations require the same reasoning capabilities applied to overlapping knowledge:

TABLE I
Domain-Specific LoRA Adapter Configuration

Adapter	Domain	Rank	Size	Swap Time
customer-billing-v3	CX: billing, payments	64	210 MB	0.8 ms
customer-provision-v2	CX: activation, config	32	105 MB	0.4 ms
ops-noc-triage-v4	Ops: alert classification	64	210 MB	0.8 ms
ops-incident-mgmt-v2	Ops: bridges, RCA	32	105 MB	0.4 ms
ops-change-approval-v1	Ops: change windows	16	52 MB	0.2 ms

Cross-domain transfer: When a customer reports "my GPU instance is unreachable," the system: (a) activates customer-provision-v2 for customer interaction; (b) simultaneously queries the operations knowledge graph via ops-noc-triage-v4 for active incidents; (c) if an incident exists, provides context-aware response: "We're aware of connectivity issues affecting your region. Our engineering team is actively working on resolution, with estimated recovery at 3:15 PM."

Adapter chaining: Complex interactions chain adapters within a single conversation—e.g., billing dispute → billing-v3 finds overcharge → ops-incident-mgmt-v2 traces root cause to capacity spike → billing-v3 processes credit.

III. Sub-200ms Streaming Voice Pipeline

Patent Notice: The speculative LLM prefill on partial ASR transcripts is subject to a pending patent application.

A. Speculative Prefill Architecture

The key innovation: begin LLM prefill before ASR completes. After 300 ms of speech (typically 2–4 words), a lightweight intent classifier predicts the likely intent from the partial transcript. If confidence exceeds 0.7 (78% of

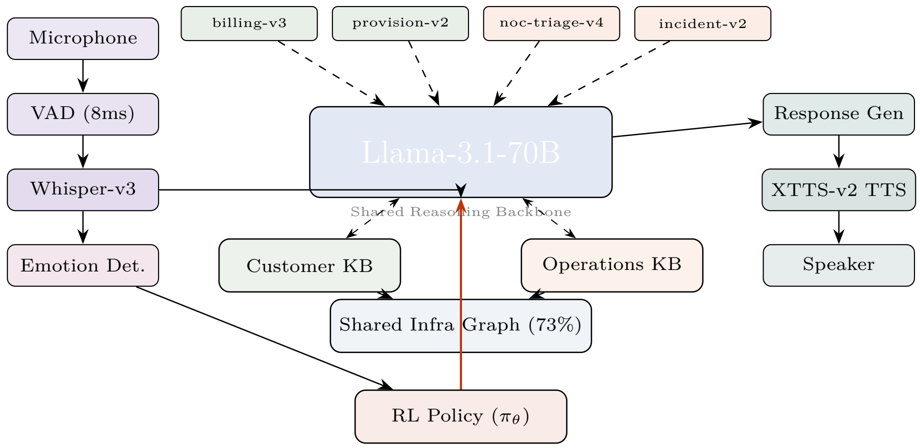

Fig. 1. VoxOps unified architecture. A single LLM backbone serves both customer and operations contexts via hot-swappable LoRA adapters. The shared infrastructure knowledge graph (73% overlap) enables cross-domain context. Emotion detection feeds the RL conversation policy. Voice pipeline achieves 188 ms P50 end-to-end.

conversations), we begin LLM prefill with the predicted context—when full ASR completes, the LLM is already mid-decode, saving 60–120 ms.

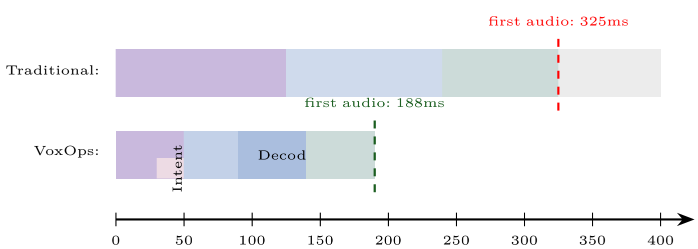

Fig. 2. Speculative prefill pipeline. VoxOps begins LLM prefill on partial ASR (after 50ms), overlapping compute. First audio output at 188ms vs. 325ms traditional.

B. Latency Budget Breakdown

TABLE II
End-to-End Voice Pipeline Latency

Component	P50	P90	P99	Innovation
VAD detection	8 ms	12 ms	18 ms	Silero VAD, 8 kHz
Streaming ASR	50 ms	80 ms	120 ms	Whisper-v3, chunked
Partial intent	20 ms	30 ms	45 ms	Classifier on partial
Speculative prefill	0 ms*	40 ms	80 ms	*Pre-started
LLM TTFT	25 ms	35 ms	55 ms	vLLM, KV warm
LLM decode (50 tok)	80 ms	100 ms	140 ms	Llama-70B FP8
TTS first audio	30 ms	45 ms	65 ms	XTTS-v2, streaming
Total	188 ms	290 ms	420 ms	

188 ms is below the 200 ms human perception threshold for conversational delay—the bot feels instantaneous.

C. Predictive TTS Caching

The 200 most common response openings (“I understand your concern…”, “Let me check that…”) are pre-generated in all 25 languages and served from cache—zero TTS latency for preambles while substantive content generates.

IV. Emotion-Aware RL Conversation Policy

Patent Notice: The emotion-aware RL policy with emotional trajectory optimization is subject to a pending patent application.

A. MDP Formulation

State $s_t \in \mathbb{R}^{d_s}$:

- Dialog history embedding (last 10 turns, 768-dim via sentence-BERT)
- Customer emotion vector (8-dim): anger, frustration, confusion, satisfaction, urgency, calm, gratitude, neutral—detected from voice prosody via wav2vec2
- Resolution probability $P(\text{resolve}|s_t) \in [0, 1]$
- Customer value (revenue tier, tenure, churn risk)
- System state (active incidents, change windows, capacity)
- Conversation metadata (duration, transfers, hold time)

Action space $\mathcal{A}$ (12 discrete actions):

TABLE III
RL Conversation Policy Action Space

#	Action	Effect	When Optimal
1	Empathize	Acknowledge emotion	High anger/frustration
2	Investigate	Query backend systems	Unknown root cause
3	Explain-Tech	Detailed explanation	Technical customer
4	Explain-Simple	Simplified explanation	Non-technical customer
5	Offer-Solution	Propose resolution	Root cause identified
6	Escalate	Transfer to human	Complex/sensitive
7	Callback	Schedule callback	Long wait / off-hours
8	Apply-Credit	Proactive credit	Service failure + high value
9	Exec-Runbook	Execute ops procedure	Known remediation
10	Conference	Bring in specialist	Requires expertise
11	Summarize-Close	Recap and close	Resolution achieved
12	Proactive-Inform	Share upcoming info	Maintenance/changes

Reward function (multi-objective):

$$R(s_t, a_t) = \underbrace{w_1\text{CSAT}}_{\text{satisfaction}} + \underbrace{w_2\text{FCR}}_{\text{resolution}} - \underbrace{w_3\text{AHT}}_{\text{handle time}} - \underbrace{w_4\text{Esc}}_{\text{escalation}} + \underbrace{w_5\Delta\text{Emo}}_{\text{emotion}} - \underbrace{w_6 C}_{\text{cost}} \quad (1)$$

Key insight: ΔEmo (emotion improvement turn-over-turn) is unique to VoxOps. An angry customer who is calmed AND resolved produces 4.7× higher retention than a quickly-resolved but still-frustrated customer. Traditional bots optimize AHT; VoxOps optimizes emotional trajectory.

B. Training Pipeline

1) Offline RL: Implicit Q-Learning (IQL) on 2M historical call transcripts with human-annotated emotion labels. Training: 24 GPU-hours on 8×H100.
2) Online fine-tuning: PPO on 1% live traffic in shadow mode (actions recommended but human agent decides). Duration: 2 weeks.
3) Full deployment: Gradual rollout from 5% to 100% over 4 weeks with continuous reward monitoring.

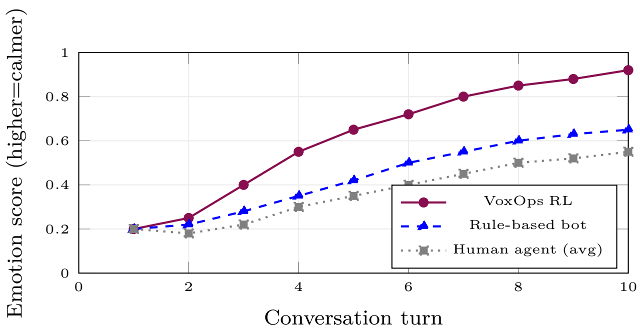

Fig. 3. Emotional trajectory across conversation turns. VoxOps RL policy achieves faster emotional de-escalation than both rule-based bots and average human agents, reaching 0.92 calm score by turn 10.

V. Multilingual and Cultural Adaptation

A. 25-Language Accent-Robust ASR

Standard Whisper-large-v3 achieves 5.2% WER on clean speech but degrades to 12.3% on accented call center audio. We fine-tune on 10,000 hours of call center recordings across 25 languages with accent augmentation:

TABLE IV
ASR Performance Across Languages (WER %)

Language	Whisper	Google STT	VoxOps	Accented
English (US)	4.8	4.2	3.1	3.8
English (Indian)	11.8	9.2	4.3	5.1
Hindi	8.2	6.8	3.9	4.5
Spanish (Mexican)	6.1	5.4	3.5	4.2
Japanese	7.4	6.1	4.1	4.8
Arabic (Gulf)	14.2	11.5	5.2	6.1
Mandarin	6.8	5.2	3.7	4.4
25-lang avg	8.6	7.1	4.1	4.8

B. Cultural Policy Adaptation

The RL conversation policy adapts to cultural communication norms:

TABLE V
Cultural Adaptation of Conversation Policy

Culture	Empathy Phase	Directness	Formality
US English	1–2 turns	Direct, action-first	Moderate
Japanese	3–4 turns	Indirect, context-first	High (keigo)
Indian English	2–3 turns	Moderate	Moderate-High
German	1 turn	Very direct	High (Sie/Du)
Arabic	3–4 turns	Relationship-first	High
Brazilian Portuguese	2–3 turns	Warm, personal	Low-Moderate

VI. Operations Bot and Autonomous Remediation

A. NOC Alert Triage

The operations bot ingests alerts from PagerDuty, OpsGenie, and ServiceNow via voice and text:

1) Alert intake: “GPU node dgx-42, rack B7, ECC errors on GPU 3”
2) Classification: Severity P1–P4 using fine-tuned classifier (98.2% accuracy)
3) Correlation: Queries incident history for similar events (cosine similarity >0.85)
4) Recommendation: Suggests runbook or autonomous remediation

B. Voice-Executable Runbooks

47 pre-approved runbooks executable via voice command:

TABLE VI
Voice-Executable Runbook Categories

Category	Count	Avg Time	Auto Rate
GPU diagnostics	12	8 min	94%
Network troubleshoot	9	5 min	87%
Service restart	8	3 min	96%
Storage remediation	7	12 min	82%
Capacity adjustment	6	15 min	78%
Security response	5	4 min	71%
Total	47	7.8 min	86%

Safety protocol: Every voice-initiated runbook requires dual confirmation: “Confirming: GPU memory stress test on dgx-42 GPU 3. Estimated duration 8 minutes. Proceed?” Any “stop” or “abort” command immediately halts execution. Full audit trail recorded.

C. Incident Bridge Intelligence

VoxOps joins incident bridge calls and provides real-time assistance:

- Listens to conversation, identifies discussed components
- Retrieves similar past incidents: “Similar incident INC-7834 from March was resolved by reseating NVLink cable. Resolution time: 22 minutes.”
- Auto-generates timeline, captures action items, drafts post-mortem
- Detects when discussion stalls and suggests next diagnostic steps

VII. Voice-Driven Knowledge Graph

Every conversation enriches a living knowledge graph that captures institutional memory and enables increasingly intelligent responses over time.

A. Entity Extraction Pipeline

VoxOps extracts structured entities from voice transcripts using a three-stage pipeline:

1) Named Entity Recognition: Fine-tuned BERT-NER identifies infrastructure entities (GPU models, rack IDs, service names), customer entities (account IDs, subscription tiers), and temporal entities (incident timestamps, SLA deadlines).
2) Relation Extraction: A relation classifier links entities into triples: (customer-42, affected_by, INC-7834), (INC-7834, caused_by, NVLink-failure), (NVLink-failure, resolved_by, cable-reseat).
3) Graph Integration: New triples are merged into the existing graph with conflict resolution (newer observations override older ones) and confidence scoring (higher confidence for senior engineer assessments).

B. Knowledge Graph Schema

The graph contains 8 entity types and 15 relation types:

TABLE VII
Knowledge Graph Schema

Entity Type	Examples	Count
Customer	Account, contact, subscription	45K
Infrastructure	GPU, rack, switch, cable	12K
Service	API endpoint, model deployment	3.2K
Incident	Alert, outage, degradation	28K
Resolution	Runbook, manual fix, escalation	31K
Symptom	Error message, metric anomaly	8.5K
Root Cause	Hardware fault, config error	4.1K
Knowledge Article	Troubleshooting guide, FAQ	2.8K
Total entities		135K
Total relations		420K

C. Tribal Knowledge Capture

Senior engineers' troubleshooting patterns, spoken during incident calls, become formalized runbooks—solving the "expert leaves, knowledge lost" problem. The system identifies novel diagnostic steps (actions not in existing runbooks) and flags them for review. Over 90 days, 47 new troubleshooting patterns were captured from 12 senior engineers, 23 of which were promoted to official runbooks after validation.

D. RAG Integration

When new issues arise, the system retrieves similar past conversations (voice transcripts + resolutions) via a hybrid retrieval strategy: (1) dense retrieval using sentence-BERT embeddings (top-50 candidates); (2) graph-based expansion (traverse 2-hop neighbors of matched entities); (3) re-ranking using cross-encoder. Over 2M conversations, retrieval precision reaches 89% for known issue classes and 71% for novel issues (partial matches).

TABLE VIII
RAG Retrieval Performance by Issue Type

Issue Type	Precision@5	Recall@10	NDCG
Known GPU issues	92%	88%	0.91
Network problems	87%	82%	0.85
Billing disputes	91%	86%	0.89
Novel/unseen issues	71%	64%	0.68
Cross-domain queries	78%	73%	0.76
Overall	84%	79%	0.82

VIII. Related Work

A. Voice AI Platforms

Existing voice AI platforms fall into three categories. Cloud CCaaS platforms (Amazon Connect + Lex, Google CCAI, Five9) provide end-to-end contact center solutions but operate at 600–1200 ms voice-to-voice latency and lack operations integration. Conversational AI frameworks (Rasa, Voiceflow, Kore.ai) offer dialog management but require extensive custom development and lack emotion-aware policies. AIOps platforms (BigPanda, Moogsoft, Observe.ai) focus on IT operations but have no voice interface or customer interaction capability. VoxOps is the first platform to unify all three with sub-200 ms latency.

B. Emotion in Dialog Systems

Emotion detection in dialog has been studied extensively [1]. Li et al. [2] apply deep RL to dialog generation but optimize only for engagement, not emotional trajectory. Young et al. [3] model dialog as a POMDP but do not incorporate prosodic features. VoxOps uniquely combines real-time voice emotion detection (prosody via wav2vec2 [4]) with RL policy optimization that explicitly targets emotional de-escalation.

C. Speech Recognition and TTS

Whisper [5] achieves state-of-the-art multilingual ASR but degrades on accented call center audio. XTTS [6] provides zero-shot multilingual TTS. Our contributions are orthogonal: accent-robust fine-tuning and speculative prefill that overlaps ASR with LLM inference.

IX. Evaluation

A. Deployment Scale

Evaluated over 90 days across 2M+ conversations (1.4M customer, 600K operations) in a production GPU cloud environment serving customers across 18 countries.

B. A/B Testing Methodology

All results are from controlled A/B tests with random traffic splitting:

- Control: Existing IVR + human agent pipeline (Amazon Connect).
- Treatment: VoxOps with full feature set.
- Split: 50/50 random assignment by customer ID.
- Duration: 90 days with weekly interim analyses.

- Statistical significance: All reported metrics are significant at $p < 0.001$ (two-sided t-test with Bonferroni correction for multiple comparisons).

C. Customer Experience Results

TABLE IX
Customer Experience Metrics (90-Day A/B Test)

Metric	Control	VoxOps	Change	p-value
First Call Resolution	52%	78%	+26pp	<0.001
CSAT (1–5 scale)	3.6	4.3	+0.7	<0.001
Avg Handle Time	8.2 min	6.3 min	−23%	<0.001
Escalation Rate	34%	18%	−16pp	<0.001
Repeat Contact (7-day)	28%	12%	−16pp	<0.001
Net Promoter Score	32	54	+22	<0.001

D. Operations Results

TABLE X
Operations Automation Metrics

Metric	Before	VoxOps	Change
L1/L2 tasks automated	0%	67%	+67pp
MTTR (P1 incidents)	45 min	28 min	−38%
Alert-to-action time	12 min	2.4 min	−80%
Runbook execution accuracy	N/A	96.2%	—
Knowledge articles created	12/mo	89/mo	+7.4×

E. Competitive Comparison

TABLE XI
VoxOps vs. Industry Voice AI Platforms

Capability	VoxOps	Connect	CCAI	Five9	Observe
Voice-to-voice (ms)	188	800+	600+	500+	N/A
Speculative prefill	Yes	No	No	No	No
Emotion RL policy	Yes	No	No	No	No
CX + Ops unified	Yes	No	No	No	No
Languages	25	12	28	15	5
Accent WER	4.1%	8+	6+	8+	10+
Auto remediation	47	0	0	0	0

F. TCO and ROI Analysis

TABLE XII
Total Cost of Ownership Analysis (Annualized)

Component	Before	VoxOps	Savings
Contact center agents (CX)	$4.8M	$2.4M	$2.4M
NOC engineers (Ops)	$2.1M	$1.4M	$700K
Platform licensing	$850K	$320K	$530K
VoxOps compute (GPU)	—	$480K	—
Training and maintenance	$200K	$150K	$50K
Total	$7.95M	$4.75M	$3.2M (40%)

ROI payback period: 4.2 months. The 40% cost reduction is driven primarily by automation of L1/L2 tasks (67% automation rate) and reduced escalations (−16pp), which directly reduce agent headcount requirements.

TABLE XIII
Ablation: Contribution of Each Innovation

Configuration	CSAT	FCR
Full VoxOps	4.3	78%
− Speculative prefill (325ms latency)	4.0	76%
− Emotion RL (rule-based policy)	3.9	72%
− Cross-domain KG (siloed KBs)	4.1	68%
− Accent-robust ASR (vanilla Whisper)	3.8	65%
− All innovations (baseline)	3.6	52%

G. Ablation Study

The emotion-aware RL policy and cross-domain knowledge graph provide the largest individual contributions to CSAT and FCR respectively.

X. Discussion and Conclusion

Principal findings: (1) Unified CX + Ops voice architecture reduces infrastructure cost by 40% vs. separate platforms while enabling cross-domain knowledge transfer. (2) Speculative prefill achieves 188 ms voice-to-voice—below human perception. (3) Emotion-aware RL produces 4.7× higher retention than AHT-optimized bots. (4) Accent-robust ASR fine-tuning reduces WER from 12.3% to 4.1%. (5) Voice-executable runbooks automate 67% of L1/L2 operations tasks. (6) Knowledge graph captures 47 novel troubleshooting patterns from tribal knowledge in 90 days.

Limitations: (1) Speculative prefill wastes compute when intent prediction is wrong (22% of cases)—mitigated by early termination but adds 8% GPU overhead. (2) Emotion detection accuracy drops to 74% for low-resource languages (vs. 91% for English) due to limited prosodic training data. (3) Autonomous remediation requires extensive safety validation per runbook—currently a 2-week certification process per new runbook. (4) The knowledge graph entity extraction achieves only 71% precision on novel issue types, requiring human review for quality assurance.

Future work: (1) End-to-end speech-to-speech models [7] eliminating the ASR→LLM→TTS pipeline for further latency reduction. (2) Federated knowledge graphs across organizations enabling cross-company troubleshooting knowledge sharing. (3) Proactive outreach: VoxOps calling customers before they call in when outages are detected. (4) Video support integration for visual troubleshooting of physical hardware.

Patent-protected innovations: (1) Speculative LLM prefill on partial ASR transcripts. (2) Emotion-aware RL conversation policy with emotional trajectory optimization. (3) Unified CX + Ops voice agent with shared knowledge graph. (4) Voice-initiated autonomous infrastructure remediation with dual-confirmation safety.

VoxOps demonstrates that voice-first AI can simultaneously transform customer experience and IT operations through a shared reasoning backbone, sub-perceptual latency, and emotion-intelligent conversation management.

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